CELL: THE UNIT OF LIFE

Let's Go..

🔬 2.1 What is a Cell?

- The cell is the fundamental structural and functional unit of all living beings.
- It is the smallest part of the body of an organism capable of independent existence.
- It is capable of performing the essential functions of life.

📜 2.3 The Cell Theory

- **The Discoverers:**
    - **Matthias Schleiden (1838):** A German Botanist who announced every plant is made of a large number of cells.
    - **Theodor Schwann (1839):** A German zoologist who declared all animals and plants are composed of cells.
- The Cell Theory was officially proposed by Schwann and Schleiden in the year **1839**.
- **The Big Addition:** Rudolf Virchow in **1858** added that all cells arise from pre-existing cells.
- **Three Major Points of Cell Theory:**
    1. The cell is the smallest unit of structure of all living things.
    2. The cell is the unit of function of all living things.
    3. All cells arise from pre-existing cells.

📊 2.4 Cells - How Many?

- The bigger an organism, the greater the number of cells in its body.

Parameter 📏	Single-Celled 🦠	Few-Celled 🌿	Multi-Celled 🌳
Cell Count	Made up of just one single cell.	A few hundred or a few thousand cells.	Millions and billions of cells.
Inline Examples	➡️ *(Example: Bacteria, yeast, amoeba).*	➡️ *(Example: Spirogyra, Volvox).*	➡️ *(Example: Human beings, Mango trees).*

- **Human Cell Statistics:**
    - Whole body: ~37.2 trillion cells.
    - Human Brain: ~100 billion nerve cells.
    - Blood (per cubic mm): 4-6 million red blood cells, 7 thousand white blood cells.

📏 2.5 Cells - How Small?

- **Smallest Cells:** Bacteria (0.3 - 5.0 micrometre) and red blood cells (~7 micrometre).
- **Longest Cells:** Nerve cells.
- **Largest Cells:** Birds' eggs ➡️ *(Example: Ostrich egg is the largest single cell).*

⚡ 2.6 & 2.7 Smallness of Cells & Shapes

- **Why are cells small? (Two Reasons):**
    1. Allows different regions to communicate with each other rapidly.
    2. Provides a large surface area/volume ratio.
- ⚙️ **Function of Large Surface Area:** Ensures greater diffusion of nutrients into the cell, metabolic wastes out, and respiratory gases (oxygen in, carbon dioxide out).
- **Human Red Blood Cells (RBCs):** Circular and biconcave shape.

    - 📍 **Location:** Narrow blood capillaries.
    - ⚙️ **Function:** To easily pass through narrow capillaries and transport oxygen.
- **White Blood Cells (WBCs):** Amoeboid shape.
    - 📍 **Location:** Blood capillaries.
    - ⚙️ **Function:** Amoeba-like movement with pseudopodia allows them to squeeze out through capillary walls.

📝 Part 1: In-Chapter Point-Wise Notes

🔬 2.8 Structure of a Cell

- A generalised cell consists of three essential parts: (1) cell membrane (plasma membrane), (2) nucleus, and (3) cytoplasm.
- Organelles ("little organs") are parts of a cell that have a definite shape, a definite structure, and a definite function.
- Cell organelles are the living parts of the cell.

🧱 2.8.1 Cell Boundaries

- Cell boundaries are outer protective structures that separate the internal contents of a cell from its external environment.
- ⚙️ **Function:** They regulate the movement of substances in and out of the cell, helping to maintain its shape and internal balance.

Parameter 📏	**Cell Membrane (Plasma Membrane)** 🟢	**Cell Wall** 🌿
Living Status	It is living.	It is a non-living substance.
Composition	Composed of lipoproteins.	Made of cellulose.
Permeability	Selective permeability (allows certain substances	Freely permeable (allows substances in solution to

Parameter 📏	Cell Membrane (Plasma Membrane) 🟢	Cell Wall 🌿
	to pass while preventing others).	enter and leave without hindrance).
Presence	Bounds each cell.	Surrounds the cell membrane in plant cells only.

💧 2.8.2 Cytoplasm & Embedded Organelles

- Cytoplasm is a semi-liquid substance that occupies most part of the cell within the cell membrane.
- Under a compound microscope, it appears to be colourless, partly transparent, and somewhat watery.
- Living cytoplasm is always in a state of some movement.
- **Endoplasmic Reticulum (ER):** An irregular network of double membranes.
    - 📍 **Location:** Distributed over the entire cytoplasm; outer end connected to cell membrane, inner end to nuclear membrane.
    - ⚙️ **Function:** Forms the supporting framework of the cell and serves as a pathway for the distribution of materials.
    - It appears rough when small granules called ribosomes are attached, and smooth without them.
- **Ribosomes:** Numerous small granules.
    - 📍 **Location:** Scattered freely in the cytoplasm or attached to the membranes of the endoplasmic reticulum.
    - ⚙️ **Function:** They are the factories for the synthesis of proteins.
- **Mitochondria:** Spherical or rod-shaped bodies, minute double-walled bags with inner walls projecting inwards called cristae.
    - They contain their own DNA which can replicate.

    - ⚙️ **Function:** Sites where cell respiration occurs to release energy. Energy is stored in the form of ATP (adenosine triphosphate). They are called the "power houses of the cell".

- **Golgi Apparatus:** Occurs in the form of granules, filaments, or rods originating from the ER. Consists of groups of hollow tubular structures with membranous walls (cisternae) and vesicles/vacuoles.
    - 📍 **Location:** Generally located near the nucleus.
    - ⚙️ **Function:** Concerned with the secretions of the cell (enzymes, hormones) and their packaging and transport.
    - In plants, Golgi complexes are known as **dictyosomes**.
- **Lysosomes:** Small vesicles of different shapes containing digestive enzymes.
    - ⚙️ **Function:** Destroy and digest foreign substances around them, and digest stored food during unfavourable conditions.
    - Called "suicide bags" because many damaged cells are rapidly destroyed or dissolved by their own lysosomes.
- **Centrosome:** Found only in an animal cell. Contains two centrioles (bundles of microfilaments at right angles).
    - ⚙️ **Function:** Spindle fibres develop from the centrosome during cell division.
- **Plastids:** Found only in plant cells. Are oval, spherical, and disc-shaped.
    - **Leucoplasts:** Colourless plastids with no pigment. ⚙️ **Function:** Store starch.
    - **Chromoplasts:** Variously coloured plastids (yellow, orange, and red). Contain pigments xanthophyll (yellow) and carotene (orange-red). 📍 **Location:** Mostly present in petals of flowers and in fruits.
    - **Chloroplasts:** Green-coloured plastids containing the pigment chlorophyll. Have DNA and the capacity to divide. ⚙️ **Function:** Trap solar energy and absorb carbon dioxide to manufacture starch and sugar during photosynthesis.

- **Non-Living Substances (Cell Inclusions):**
    - **Granules:** Small particles in the cytoplasm containing food materials like starch, glycogen, and fats.
    - **Vacuoles:** Clear spaces filled with water and substances in solution called cell sap. Plant cells have large vacuoles; animal cells have smaller ones.

🧬 2.8.3 Nucleus

- The nucleus is the most important part of the cell.
- ⚙️ **Function:** It regulates and coordinates various life processes, plays an important part in cell division, and contains factors (genes) which determine heredity.
- **Structure:** It is a small spherical mass with a delicate nuclear membrane, filled with dense nucleoplasm.
    - **Chromatin fibres:** Thread-like structures in the nucleoplasm. During cell division, they become thick and ribbon-like and are called **chromosomes**.
    - **Nucleolus:** Participates in protein synthesis of ribosomes.
- **Chromosomes & Genes:**
    - Chromosome number is definite in each species. Human body cell has 46 (23 pairs).
    - Chromosomes carry genetic characters from parents to offspring.
    - Chromosomes are made of chromatin, which is composed of hereditary units called **genes**.
    - Genes are made of DNA (deoxyribonucleic acid).
    - It is the genes, not the number of chromosomes, that determine the characteristics of a species.

Parameter 📏	Prokaryotic Cells 🦠	Eukaryotic Cells 🐅
Nuclear Membrane	Nuclear membrane is absent.	Double nuclear membrane is present.
Nuclear Material	Chromatin fibres occur freely in the central region of the cytoplasm called nucleoid.	Contained within the true nucleus.
Inline Examples	➡️ *(Example: Bacteria).*	➡️ *(Example: All organisms other than bacteria).*

📝 Part 1: In-Chapter Point-Wise Notes

🧱 2.9 The Plant and Animal Cells

- Both plant and animal cells share basic structures like the plasma membrane, cytoplasm, nucleus, endoplasmic reticulum, golgi apparatus, mitochondria, and ribosomes.

Parameter 📏	Plant Cells 🌿	Animal Cells 🐅
Cell Wall	A definite cell wall is present, made up of cellulose.	No cell wall is present.
Centrosome	Centrosome is not present.	Centrosome is present.
Vacuoles	Vacuoles are prominent, one or more.	Vacuoles, if any, are small and temporary.

Parameter 🪒	Plant Cells 🌿	Animal Cells 🐅
Plastids	Usually contain plastids.	Do not contain plastids.
Size & Outline	Usually larger, with distinct outlines.	Usually smaller, with less distinct boundaries.
Cytoplasm Density	Cytoplasm is not so dense.	Cytoplasm is denser and more granular.
Cytoplasm Arrangement	Mostly pushed to the periphery as a thin lining.	Fills almost the entire cell.

🧪 2.10 Protoplasm

- **Formula:** Cytoplasm + Nucleus = Protoplasm.
- It is the living substance in an organism, described as a translucent fluid that is somewhat colourless, greyish, or brownish.
- **Composition:** Contains carbon, hydrogen, oxygen, nitrogen, sulphur, iron, and phosphorus in the form of specific compounds (water, proteins, carbohydrates, fats, mineral salts).
- It is impossible to make an accurate chemical analysis of protoplasm because it ceases to be protoplasm the moment it is removed from the organism.

🔬 Definition of the Cell (Recap)

- The basic structural building block of living organisms consisting of protoplasm (living substance) enclosed by a cell membrane, with an additional cell wall in plant cells, and having a nuclear membrane or without it.

🧬 2.11 Prokaryotic and Eukaryotic Cells

- **Prokaryotic:** Refers to cells containing a primitive nucleus (pro = early/primitive; karyon = nucleus).

- **Eukaryotic:** Means cells with a perfectly formed nucleus (eu = true; karyon = nucleus).

Parameter 📏	Prokaryotic Cell 🦠	Eukaryotic Cell 🐅
Nucleus Structure	Nucleus is not well defined (called nucleoid).	Well defined nucleus with a nuclear membrane.
Genetic Material	A single length of only deoxyribonucleic acid (DNA).	Several lengths of genetic material (chromosomes) containing DNA wound around certain proteins.
Ribosome Size	Contains small ribosomes only.	Contains larger ribosomes.
Other Organelles	No other cell organelles present.	Several organelles like mitochondria, endoplasmic reticulum, chloroplasts, etc..
Inline Examples	➡️ *(Example: Bacteria, Blue green algae (Cyanobacteria)).*	➡️ *(Example: Euglena, Amoeba, all plants and animals).*

⚙️ 2.12 Every Activity of a Living Organism is the Outcome of Cellular Activity

- There is not a single activity in the body that is not carried out by cells.
- **Growth:** Due to the growth in size and increase in the number of cells.
- **Repair:** Regeneration of a lost part (e.g., lizard tail) is due to cell divisions.

- **Movement:** Bending, drooping, or locomotion is due to the contractility of cells (e.g., muscle cells in animals, changes in water content in plant leaves).
- **Feeding & Nutrition:** Involves sensory cells (tasting), muscle cells (chewing), gland cells (secreting digestive enzymes), and cells of the intestine lining (absorbing food).
- **Circulation:** Blood flow is propelled by the contraction of muscle cells in the heart.
- **Respiration:** Respiratory gases are transported by blood cells.
- **Protection:** White Blood Cells (W.B.C.) devour disease-causing germs or produce antibodies and antitoxins.
- **Thermoregulation (Heat Maintenance):** Maintained by cellular activity, and cooled by sweating from skin sweat glands.
- **Plant-Specific Activities:** Absorption of water/nutrients is via root cells; light trapping for food is via chloroplasts in leaf cells; attracting insects for pollination is via colour in petal cells or nectar-secreting cells.
- **Inheritance:** Transmission of parental features to young ones depends on germ cells (egg and sperm).

🏆 Part 2: End-of-Chapter Master Lists (The Grand Finale)

📚 1. Combined Terms List (Glossary)

Biological Term 🔑	Exam-Ready Definition 🧠
Cell	The fundamental **structural and functional** unit of all living beings.
Cell Theory	The biological principle stating that cells are the basic units of structure and function, and all cells arise from **pre-existing cells**.

Biological Term 🔑	Exam-Ready Definition 🧠
Protoplasm	The total **living substance** of a cell, consisting of the cytoplasm and the nucleus.
Pseudopodia	False feet utilized by white blood cells for **amoeboid movement** to squeeze through capillary walls.
Organelles	Living parts of a cell that have a definite shape, structure, and perform **specific functions**.
Cristae	The finger-like processes projecting inwards from the inner walls of **mitochondria**.
ATP	Adenosine triphosphate, an **energy-rich compound** produced and stored in mitochondria.
Dictyosomes	Golgi complexes and their more scattered groups of cisternae and vesicles found specifically in **plants**.
Cisternae	Hollow, tubular structures with membranous walls found in the **Golgi apparatus**.
Tonoplast	The covering that encloses a **vacuole** in a plant cell.
Anthocyanins	Purple and blue colouring pigments dissolved in the **cell sap** of vacuoles, not associated with plastids.
Nucleoid	The central region of the cytoplasm in **prokaryotic cells** where the single length of DNA occurs freely.

Biological Term 🔑	Exam-Ready Definition 🧠
Genes	Hereditary units composed of **DNA** that make up chromatin and determine the characteristics of a species.
Prokaryotic Cell	Cells containing a primitive, not well-defined nucleus (nucleoid) lacking a **nuclear membrane**.
Eukaryotic Cell	Cells containing a perfectly formed, well-defined nucleus with a **double nuclear membrane**.

🌟 2. Master Example List

- **Organisms by Cell Count:**
  - *Single-celled:* Bacteria, yeast, amoeba.
  - *Few-celled:* Spirogyra, Volvox.
  - *Multi-celled:* Human beings, Mango trees.
- **Cell Size Extremes:**
  - *Smallest:* Bacteria, Red blood cells.
  - *Longest:* Nerve cells.
  - *Largest:* Ostrich egg.
- **Specific Cell Structures & Occurrences:**
  - *Cell walls of dead cells:* Cotton, jute, coconut fibres.
  - *Leucoplasts (Starch Storage):* Cells of a potato.
  - *Chromoplasts (Pigments):* Petals of flowers and in fruits.
  - *Prokaryotic Organisms:* Bacteria, Blue-green algae (Cyanobacteria).
  - *Eukaryotic Organisms:* Euglena, Amoeba, all plants and animals.
- **Cellular Activities Manifested in Organisms:**

- *Repair/Regeneration:* Tail of a lizard.
- *Protection:* White Blood Cells (devour germs).
- *Thermoregulation (Cooling):* Sweat from sweat glands in the skin.
- *Absorption:* Root cells in plants.
- *Inheritance:* Germ cells (egg and sperm).

⚙️ 3. Master Function & Location Matrix

Biological Part 🧬	**Exact Anatomical Location** 📍	**Primary Function** ⚙️
Red Blood Cells (RBCs)	Inside narrow blood capillaries.	Pass through capillaries and transport oxygen.
White Blood Cells (WBCs)	Inside and outside blood capillaries.	Use amoeboid movement to squeeze out through capillary walls; devour germs and provide protection.
Nerve Cells	Extending from fingertips up to the spinal cord.	Rapid communication between different body regions.
Cell Wall	Outermost in plant cells (surrounding plasma membrane).	Gives rigidity and shape; allows substances to enter/leave freely; provides protection.
Plasma Membrane (Cell Membrane)	Outermost in animal cells; lies next to cell wall in plant cells.	Separates contents from surroundings; regulates entry of certain solutes and ions; maintains cell shape in animals.

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function ⚙️
Cytoplasm	Inside the plasma membrane, excluding the nucleus.	Medium for organelles to perform functions; site of all metabolic activities; medium for initial steps of respiration (pyruvic acid).
Endoplasmic Reticulum (ER)	Continuous with plasma membrane on the outside and nuclear membrane on the inside.	Provides supportive framework for the cell; synthesizes and transports proteins and fats.
Mitochondria	Distributed throughout the cytoplasm.	Release of energy from pyruvic acid in the form of ATP; seat of cellular aerobic respiration; synthesis of respiratory enzymes.
Golgi Apparatus (Dictyosome)	Generally located near the nucleus.	Synthesis and secretion of enzymes, hormones, etc.; formation of acrosome of sperm (in animals/humans).
Ribosomes	Scattered in cytoplasm or attached to the outside of the endoplasmic reticulum.	Protein synthesis.
Lysosomes	Budded off from Golgi bodies into the cytoplasm.	Intracellular digestion; destroy foreign substances; rapidly destroy old/injured organelles ("suicide bags");

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function ⚙️
		digest cartilages during bone formation.
Centrosome	A region surrounding centrioles, located near the nucleus (Animal cells only).	Initiates and regulates cell division; forms spindle fibres with the help of asters.
Chromoplasts (Plastids)	Petals of flowers and in fruits (Plant cells only).	Impart colour (yellow, orange, red) to help in pollination of the flower and dispersal of seeds.
Chloroplasts (Plastids)	Leaves and green parts of plants (Plant cells only).	Trap solar energy for photosynthesis; contain green pigment chlorophyll.
Leucoplasts (Plastids)	Roots, underground stems (e.g., potato) (Plant cells only).	Store starch (contains no pigments).
Nucleus	Mostly spherical and dense, located within the cytoplasm.	Regulates cell functions; contains chromosomes (bearers of genes that control hereditary characters).
Nucleolus	One or more round-shaped substructures	Produces ribosomes; participates in protein synthesis by forming and

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function ⚙️
	present inside the nucleus.	storing RNA; dictates ribosomes to synthesise proteins.
Chromatin Fibres	Inside the nucleoplasm during the resting stage.	Condense into chromosomes during cell division to carry hereditary information/genes from parents to offspring.
Vacuoles	Suspended in the cytoplasm (larger in plants, smaller/temporary in animals).	Store water, food, and waste products; give turgidity to plant cells; contain pigments like anthocyanins.
Granules	Suspended in the cytoplasm.	Store starch (in plant cells), glycogen (in animal cells), and fat-containing globules which serve as food for the cell.
Biological Term 🔑	**Exam-Ready Definition 🧠**	
Cell Inclusions	The **non-living** parts of the cell, specifically vacuoles and granules, which are suspended in the cytoplasm and are important in their own way.	

Here are all the questions and their perfect 100/100 answers, formulated strictly based on the notes and the provided image text! 🚀 💯 I have written out every question before providing the answer, exactly as you requested. ✨

📘 C. SHORT ANSWER TYPE

1. Define the terms:

- **(a) Cell organelles:** Parts of a cell that have a definite shape, a definite structure, and a definite function. They are the living parts of the cell.
- **(b) Cell inclusions:** The non-living parts of the cell, specifically vacuoles and granules, which are suspended in the cytoplasm.
- **(c) Cytosol:** A jelly-like fluid that surrounds the organelles.
- **(d) Cell sap:** The water and substances in solution that fill the clear spaces called vacuoles.
- **(e) Cytoplasm:** A semi-liquid substance that occupies most part of the cell within the cell membrane.

2. Write the exact locations of each of the following:

- **(a) Centrosome:** Located near the nucleus in animal cells.
- **(b) Chromosome:** Found inside the nucleus (nucleoplasm).
- **(c) Genes:** Located on the chromosomes within the nucleus.
- **(d) Mesosome:** Located as an inward fold of the cell membrane in prokaryotic cells.
- **(e) Plasmid:** Found within the cytoplasm of a prokaryotic cell as a single length of DNA.

3. Mention three features found only in plant cells and one found only in animal cells.

- **Only in plant cells:** A definite cell wall made of cellulose, large prominent vacuoles, and plastids.
- **Only in animal cells:** Centrosome.

4. Why are the cells generally of a small size?

- Cells are small to allow different regions of the cell to communicate with each other rapidly.
- Small size provides a large surface area/volume ratio for greater diffusion of nutrients, metabolic wastes, and respiratory gases.

5. It is said that the protoplasm cannot be analysed chemically. Why?

- Because it ceases to be protoplasm (it loses its living properties) the moment it is removed from the organism for chemical analysis.

6. What is the difference between an organ and an organelle?

- An **organ** is a large, macroscopic body part made of tissues that performs specific functions for the whole organism.
- An **organelle** is a microscopic, living "little organ" strictly within a cell that has a definite structure and function.

7. Do you think the cells of an elephant would be larger than the cells of a rat? Explain briefly.

- No, the cells would not be larger. The size of an organism depends on the *number* of cells in its body, not the size of the individual cells.

📕 D. DESCRIPTIVE ANSWER TYPE

1. What is the cell theory? Who propounded it and when?

- **Cell Theory:** The cell is the smallest unit of structure and function of all living things, and all cells arise from pre-existing cells.
- **Propounded by:** It was proposed by Matthias Schleiden (1838) and Theodor Schwann (1839). Rudolf Virchow added to it later in 1858.

2. Differentiate between the following pairs of terms:

- **(a) Protoplasm and cytoplasm:** Protoplasm is the *total* living substance of a cell (Cytoplasm + Nucleus). Cytoplasm is just the semi-liquid substance outside the nucleus but within the cell membrane.
- **(b) Nucleolus and nucleus:** The nucleus is the large spherical organelle that regulates cell functions and contains chromosomes. The nucleolus is a

smaller, round-shaped substructure *inside* the nucleus that produces ribosomes.

- **(c) Centrosome and chromosome:** Centrosomes are found near the nucleus in animal cells and regulate cell division. Chromosomes are thread-like structures inside the nucleus that carry hereditary information (genes).
- **(d) Cell wall and cell membrane:** The cell wall is an outer, non-living, rigid layer of cellulose found in plant cells that is freely permeable. The cell membrane is a living, lipoprotein layer found in all cells that is selectively semi-permeable.
- **(e) Plant cell and animal cell:** Plant cells contain a definite cell wall, plastids, large prominent vacuoles, and lack a centrosome. Animal cells have no cell wall, no plastids, small/temporary vacuoles, and possess a centrosome.
- **(f) Prokaryotes and eukaryotes:** Prokaryotes have a primitive, undefined nucleus (nucleoid) lacking a membrane, a single length of DNA, and small ribosomes. Eukaryotes have a perfectly formed nucleus with a double membrane, multiple chromosomes, larger ribosomes, and multiple organelles.

3. State the major functions of the following:

- **(a) Plasma membrane:** Regulates the entry of certain solutes and ions.
- **(b) Ribosome:** Protein synthesis.
- **(c) Lysosome:** Intracellular digestion; destroying foreign substances and old organelles ("suicide bags").
- **(d) Mitochondria:** Release of energy from pyruvic acid in the form of ATP; seat of cellular aerobic respiration.
- **(e) Golgi apparatus:** Synthesis and secretion of enzymes, hormones, etc.
- **(f) Cytoplasm:** Medium for metabolic activities and the initial steps of respiration.
- **(g) Asters of centrosome:** Initiate and regulate cell division by forming spindle fibres.

- **(h) Chromosomes:** Carry hereditary information or genes from parents to offspring.
- **(i) Glycogen granule:** Store food for the cell in animal cells.
- **(j) Vacuoles:** Store water and waste products, and give turgidity to plant cells.

4. Mention any three differences between a living cell and a brick in a wall.

- A living cell is made of living protoplasm, whereas a brick is non-living.
- A cell can divide and reproduce, whereas a brick cannot.
- A cell performs essential metabolic functions, whereas a brick has no metabolic functions.

5. List any six features found both in plant and animal cells.

- 1. Plasma membrane, 2. Cytoplasm, 3. Nucleus, 4. Endoplasmic Reticulum (ER), 5. Golgi apparatus, 6. Mitochondria.

🟢 E. STRUCTURED/APPLICATION/SKILL TYPE

1. Given below are the sketches of two types of cells A and B.

- **(a) Which one of these is a plant cell? Give reason in support of your answer.**
  - Cell B is a plant cell. Reason: It has a rigid cell wall, a distinct geometric shape, and large central vacuoles.
- **(b) List the cell structures which are common to both the types.**
  - Plasma membrane, nucleus, cytoplasm, mitochondria, endoplasmic reticulum, and Golgi bodies.
- **(c) Name the structures found only in plant cells and those found only in animal cells.**
  - **Only in plant cells:** Cell wall, plastids, large vacuoles.
  - **Only in animal cells:** Centrosome.

2. Gautam observed a plant cell and an animal cell under electron microscope and found one organelle that is present only in animal cells but not in plant cells. The figure of this organelle is shown ahead.

- **(a) What is this structure called?**
    - Centrosome.
- **(b) It contains one or two dot-like dense bodies, surrounded by microtubules. Name them.**
    - Centrioles.
- **(c) Write one important role of the above mentioned structure.**
    - It initiates and regulates cell division by forming spindle fibres.

3. Study the diagram given below and answer the questions that follow:

- **(a) Identify the unicellular organism shown above.**
    - Bacteria.
- **(b) Is it a prokaryotic or eukaryotic cell?**
    - Prokaryotic cell.
- **(c) Name the spherical bodies composed mainly of RNA, shown by (A).**
    - Ribosomes.
- **(d) Mention one difference between the cellular structures shown by (B) and (C) with respect to permeability.**
    - Structure (B) is the cell membrane, which is selectively semi-permeable. Structure (C) is the cell wall, which is freely permeable.
- **(e) Write one important role of the structure shown by (D).**
    - Structure (D) is the flagellum, which aids in movement.

🧩 F. ASSERTION - REASON TYPE QUESTIONS

(Codes to use for answers: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.)

(i) Assertion (A): Ribosomes are considered as the 'delivery system of the cell'.

Reason (R): Delivery system is concerned with the secretions of the cell including enzymes, hormones, etc.

- **Answer:** (d) A is false but R is true. *(Note: The Golgi apparatus, not ribosomes, is the delivery system concerned with secretions).*

(ii) Assertion (A): Leukoplasts are colourless plastids as they have no pigments.

Reason (R): Leukoplasts are normally found in the non-photosynthetic parts of a plant.

- **Answer:** (a) Both A and R are true and R is the correct explanation of A.

(iii) Assertion (A): Cristae are the finger-like projections arising from the inner walls of mitochondria.

Reason (R): These projections increase the surface area of the inner membrane and help in the production of ATP.

- **Answer:** (a) Both A and R are true and R is the correct explanation of A.

(iv) Assertion (A): Euglena is an example of a unicellular, eukaryotic organism.

Reason (R): This organism is microscopic and has a well-defined nucleus.

- **Answer:** (a) Both A and R are true and R is the correct explanation of A.

(v) Assertion (A): Chromosomes carry hereditary information from parents to offspring.

Reason (R): Chromosomes are suspended in the cytoplasm of each cell whether it is a plant cell or an animal cell. Chromosomes are composed of RNA threads.

- **Answer:** (c) A is true but R is false. *(Note: Chromosomes are in the nucleus and are composed of DNA).*

🔬 G. CASE STUDY BASED QUESTION

1. Abhay, a 9th grade student, is conducting a project for his school science fair...

- **(a) Identify the organelle which is located near the nucleus and contains two short bundles of microfilaments arranged at right angles to each other.**
    - Centrosome.
- **(b) Which organelle is considered as the 'delivery system of the cell'?**
    - Golgi apparatus.
- **(c) Cell membrane is composed of what material?**
    - Lipoproteins.
- **(d) Give two examples of cell inclusions.**
    - Granules and vacuoles.
- **(e) Name the organelle which was seen by Abhay earlier in plant cells but lacking here in animal cells.**
    - Plastids.